PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 3 · Version A

20 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ m/s^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A mechanic pulls on the end of a 0.25 m wrench with an 80 N force directed at 60° to the wrench handle.

1) What is the magnitude of the torque on the bolt? **MC5**

- **a.** 17.3 N·m
- **b.** 10 N·m
- **c.** 320 N·m
- **d.** 20 N·m
- **e.** 34.6 N·m

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1.0 kg block hangs from a light rope wrapped around a pulley that is a uniform solid disk (M = 2.0 kg, R = 0.10 m). The pulley turns without friction, and the block is released from rest.

2) What is the acceleration of the block? **MC5**

- **a.** $4.91\ m/s^2$
- **b.** $3.27\ m/s^2$
- **c.** $6.54\ m/s^2$
- **d.** $9.81\ m/s^2$
- **e.** $2.45\ m/s^2$

3) What is the tension in the rope? **MC5**

- **a.** 3.27 N
- **b.** 6.54 N
- **c.** 4.91 N
- **d.** 9.81 N
- **e.** 0 N

4) If the pulley were a thin ring with the same mass and radius, the block's acceleration would be **MC3**

a. smaller.
b. larger.
c. the same.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A solid sphere (m = 2.0 kg, R = 0.10 m) rolls without slipping from rest down an incline. It descends a vertical height of 1.5 m.

5) What is the speed of the sphere at the bottom? **MC5**

a. 5.42 m/s
b. 4.58 m/s
c. 4.43 m/s
d. 6.46 m/s
e. 3.84 m/s

6) What fraction of the sphere's kinetic energy at the bottom is rotational? **MC5**

a. 50.0%
b. 33.3%
c. 28.6%
d. 71.4%
e. 40.0%

7) How much work does static friction do on the sphere while it rolls? **MC3**

a. Zero
b. A positive amount
c. A negative amount

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A uniform 6.0 m plank of mass 30 kg rests on supports at its two ends. A 70 kg person stands 1.5 m from the left end.

8) What upward force does the right support exert? **MC5**

a. 319 N
b. 981 N
c. 662 N
d. 490 N
e. 343 N

9) What upward force does the left support exert? **MC5**

a. 981 N
b. 319 N
c. 662 N
d. 638 N
e. 490 N

10) As the person walks toward the right end, the force from the left support **MC3**

a. stays the same.
b. decreases.
c. increases.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A figure skater spins at 2.0 rev/s with her arms out ($I = 4.0\ kg{\cdot}m^2$). She pulls her arms in, reducing her moment of inertia to $1.6\ kg{\cdot}m^2$.

11) What is her new rate of spin? **MC5**

a. 12.5 rev/s
b. 0.8 rev/s
c. 2 rev/s
d. 3.16 rev/s
e. 5 rev/s

12) Her rotational kinetic energy **MC3**

a. decreases.
b. increases.
c. stays the same.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 30 kg child runs at 4.0 m/s along a line tangent to the edge of a merry-go-round and jumps on at its edge. The merry-go-round is a uniform disk of mass 120 kg and radius 2.0 m, initially at rest.

13) What is the angular speed of the merry-go-round just after the child lands? **MC5**

a. 1 rad/s
b. 2 rad/s
c. 0.667 rad/s
d. 0.333 rad/s
e. 0.4 rad/s

14) Compared with before the jump, the total kinetic energy afterward is **MC3**

a. more.
b. less.
c. the same.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A constant torque of 6.0 N·m acts on a wheel (I = 1.5 kg·m²) that starts at rest.

15) What is the angular momentum of the wheel after 4.0 s? **MC5**

a. 36 kg·m²/s
b. 9 kg·m²/s
c. 4 kg·m²/s
d. 16 kg·m²/s
e. 24 kg·m²/s

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.80 kg block on a horizontal frictionless surface is attached to a spring with k = 200 N/m. It oscillates with an amplitude of 0.10 m.

16) What is the period of the oscillation? **MC5**

a. 0.0636 s
b. 2.52 s
c. 0.199 s
d. 0.397 s
e. 15.8 s

17) What is the maximum speed of the block? **MC5**

a. 15.8 m/s
b. 2.5 m/s
c. 1 m/s
d. 1.58 m/s
e. 0.158 m/s

18) When the block is at x = A/2, its speed is what fraction of v_max? **MC3**

a. 0.71
b. 0.50
c. 0.87

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A simple pendulum has a period of 2.0 s on Earth. It is taken to a planet where g is one quarter of its value on Earth.

19) What is its period on the planet? **MC5**

- **a.** 1 s
- **b.** 0.5 s
- **c.** 4 s
- **d.** 2.83 s
- **e.** 8 s

20) If the amplitude of the swing were doubled (still small), the period would **MC3**

- **a.** stay the same.
- **b.** double.
- **c.** increase by √2.
